

Week 1: Course Orientation

Scripting Fundamentals and Introduction to AI Workflows

NMTAFE - Certificate IV Data Science and AI

Building the foundation for PyTorch and cloud ML engineering

Learning Objectives

- Understand course structure, expectations, and industry context in ML engineering
- Review foundational Python programming concepts for ML applications
- Recognize how scripting skills support end-to-end AI workflows
- Demonstrate a complete mini ML workflow using Python scripting

Course Overview: Your Journey Ahead

20-Week Program Structure:

- **Weeks 1-4:** Scripting foundations (Python, Bash, Git)
- **Weeks 5-6:** PyTorch fundamentals (tensors, autograd, neural networks)
- **Weeks 7-9:** Azure cloud infrastructure and data management
- **Weeks 10-12:** Portfolio development and production workflows
- **Weeks 13-18:** GPU computing and distributed ML training
- **Weeks 19-20:** Assessment and career preparation

Industry Context: Why These Skills Matter

Modern AI/ML Engineering Roles Require:

- **Automation:** Scripts reduce manual errors and save time
- **Reproducibility:** Version-controlled workflows ensure consistent results
- **Scalability:** Cloud deployment and distributed computing
- **Collaboration:** Documented code that teams can understand and maintain

"Every ML engineer must fluently create and modify Python scripts" - Industry standard

Real-World Application Examples

Government Agency: Automates student performance analytics

- Python scripts process data from multiple sources
- Bash automation handles file transfers and backups
- Results deployed to Azure for stakeholder access

Tech Startup: ML model deployment pipeline

- Scripts manage data preprocessing and model training
- Cloud automation scales resources based on demand
- Version control tracks all changes for audit compliance

Python Scripting Essentials for ML

Core Skills You'll Master:

- **Data handling:** Lists, dictionaries, file I/O operations
- **Function design:** Reusable, documented code modules
- **Control flow:** Conditional logic and iteration patterns
- **Library integration:** pandas, numpy, PyTorch ecosystem

Industry Best Practice: Every script should be readable, reusable, and documented

Code Example: Python Function Basics

```
def greet_student ( name, course="ML Engineering" ) :  
    """  
    Welcome function with documentation.  
    Args:  
        name (str): Student name  
        course (str): Course title  
    Returns:  
        str: Formatted welcome message  
    """  
    return f"Hello, {name}! Welcome to {course}."  
  
# Usage example  
message = greet_student( "Alex" )  
print( message )
```

Key principle: Always include docstrings and clear variable names

Hands-On Demo: Mini ML Workflow

Today's Practical Exercise:

1. **Data Loading** - Import CSV data using pandas
2. **Processing** - Apply simple classification logic
3. **Output** - Save results and generate summary

Real-world relevance: This pattern scales to complex ML pipelines with PyTorch models, cloud data sources, and automated deployment

Demo Code: Data Loading and Processing

```
import pandas as pd

# Step 1: Load sample data
data = {'student': ['Alice', 'Bob', 'Charlie'],
        'score': [85, 90, 78]}
df = pd.DataFrame(data)

# Step 2: Apply model logic (classification stub)
def classify_performance(score):
    return 'pass' if score >= 80 else 'fail'

df['result'] = df['score'].apply(classify_performance)

# Step 3: Output results
df.to_csv('results.csv', index=False)
print("Summary:", df['result'].value_counts())
```

From Simple Scripts to Production ML

Today's Exercise → Industry Reality:

Today	Production ML Pipeline
pandas DataFrame	PyTorch DataLoader
Simple if/else logic	Neural network models
CSV file output	Cloud database storage
Manual execution	Automated CI/CD deployment

The scripting patterns you learn today will scale to enterprise ML systems

Industry Tools and Technologies Preview

Programming Languages:

- **Python:** Core ML development, PyTorch, data processing
- **Bash:** Cloud automation, file management, system operations

Platforms and Services:

- **Azure:** Cloud compute, storage, managed services
- **Git/GitHub:** Version control and collaboration
- **PyTorch:** Deep learning framework and ecosystem

Assessment Overview

Portfolio-Based Assessment Approach:

- **Week 6:** ML Scripting Fundamentals Portfolio
- **Week 14:** Cloud Infrastructure Configuration
- **Week 16:** Single-GPU PyTorch Application
- **Week 18:** Distributed ML Pipeline with Industry Handover

Each assessment builds on previous skills - start strong with good scripting habits

Professional Development Pathways

Career Opportunities:

- **AI Developer:** Build and deploy ML applications
- **ML Engineer:** Design production ML systems
- **Cloud Engineer:** Manage scalable AI infrastructure
- **Data Scientist:** Develop insights from data using ML

Salary Range (Australia): \$70k - \$150k+ depending on experience and specialization

Best Practices: Professional Coding Standards

Documentation Requirements:

- Clear comments explaining code purpose
- Function docstrings with parameters and return values
- README files describing setup and usage

Code Quality Standards:

- Meaningful variable and function names
- Consistent indentation and formatting
- Error handling and validation
- Version control with descriptive commit messages

Practice Activity: Extend the Workflow

Your Challenge:

1. Modify the grade classification to use A/B/C/D grades
2. Add error handling for invalid scores
3. Create a function that finds top and bottom performers
4. Add comments and docstrings to all functions

Industry Connection: These improvements mirror real-world code review requirements

Troubleshooting and Support

Common Issues and Solutions:

- **Import errors:** Check library installation with `pip list`
- **File path problems:** Use relative paths or `os.path.join()`
- **Data format issues:** Validate input data before processing

Getting Help:

- Course LMS discussion forums
- Instructor office hours
- Peer study groups and code review sessions

Key Takeaways

- ✓ **Scripting is foundational** to all ML engineering roles
- ✓ **Documentation and standards** are industry requirements, not optional
- ✓ **Simple patterns** scale to complex production systems
- ✓ **Practice and iteration** build professional competency

"Strong scripting habits form the backbone of all advanced AI and ML systems"

Next Week Preview

Week 2: Bash Scripting and Shell Environments

- Linux/Unix command line essentials
- Automating file operations and system tasks
- Integration with cloud environments
- Building scripts that work with Python workflows

Preparation: Install Ubuntu or enable WSL on Windows

Questions and Discussion

Reflection Questions:

- Why is scripting considered foundational to ML engineering?
- How might automating workflows reduce errors and improve team productivity?
- What aspects of today's demo would you like to explore further?

Next Steps: Complete the practice activity and prepare your development environment for next week

